



Dynamic, real-time HVAC optimisation
- delivering energy savings automatically.



University of
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The Scottish Oceans Institute

Dynamic HVAC optimisation case study

£14.4k	158 MWh	29.3 tCO ₂ e
Measured gross financial benefit Approx. £10k net after c£4k annual subscription	Measured energy savings April 2025 - April 2026	Measured operational carbon reduction April 2025 - April 2026
Annualised performance at time of reporting: 207 MWh/year £19.4k/year gross benefit approx. £15k/year net after annual subscription 38.4 tCO ₂ e/year		
21%	9%	
HVAC electricity reduction	Heat reduction	
April 2025 - April 2026 weather-normalised baseline	April 2025 - April 2026 weather-normalised baseline	

Executive summary

The University of St Andrews is demonstrating how artificial intelligence can deliver measurable energy performance improvements in already well-performing buildings.

At the Scottish Oceans Institute (Gatty Marine Laboratory), **EcoAI's myCoreAI platform was deployed** as an AI-driven optimisation layer to support day-to-day HVAC optimisation and improve building performance.

Across the first April-to-April reporting period, myCoreAI delivered a **21% reduction in HVAC electricity consumption** and a **9% reduction in heat consumption**, measured against a **weather-normalised baseline**.

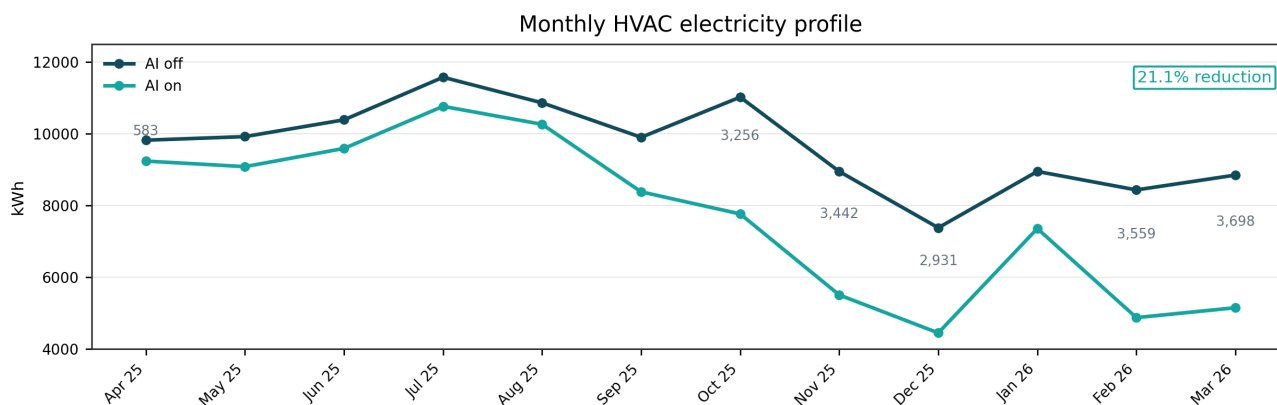
Measured energy savings over the period total approximately 158 MWh, comprising 25,896 kWh of HVAC electricity reduction and 132,218 kWh of heat reduction. This equates to approximately £14.4k of gross financial benefit, or approximately £10.4k net benefit after the annual subscription of c£4k.

Performance strengthened as myCoreAI learned the building and full control was established. At the time of reporting, annualised performance was equivalent to approximately 207 MWh/year, delivering around £19.4k/year of gross financial benefit and approximately £15k/year net benefit after annual subscription.

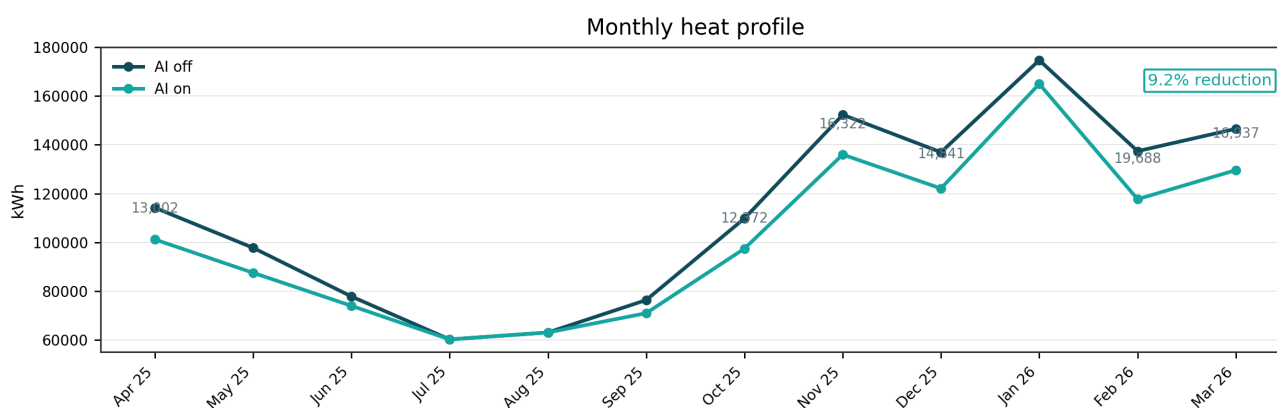
Energy savings were delivered without sacrificing comfort or disrupting building operations.

Monthly AI on vs AI off profile

The charts below show monthly weather-normalised AI off and AI on values across the April 2025 - April 2026 reporting period. For a like-for-like visual comparison, the chart view shows complete months only; April 2026 month-to-date remains included in the result table below.



Monthly HVAC electricity profile showing weather-normalised AI off and AI on performance.



Monthly heat profile showing weather-normalised AI off and AI on performance.

Note: The January 2026 data includes a short period when AI optimisation was temporarily disabled, which provided an additional practical reference point alongside the weather-normalised analysis.

Result	April 2025 - April 2026 position
HVAC electricity	AI off: 122,937 kWh AI on: 97,042 kWh Saving: 25,896 kWh (21.1%)
Heat	AI off: 1,434,961 kWh AI on: 1,302,742 kWh Saving: 132,218 kWh (9.2%)
Total measured energy saving	158,114 kWh saved
Measured gross benefit	Approx. £14.4k over April 2025 - April 2026
Measured net benefit	Approx. £10.4k after c£4k annual subscription

Background and deployment

The Scottish Oceans Institute was selected as a live reference site to assess whether AI-driven optimisation could deliver measurable additional savings in an already well-performing building.

The building operates with an established Siemens Desigo CC Building Management System (BMS), providing a strong baseline from which to evaluate whether further improvement could be achieved without replacing existing infrastructure.

myCoreAI was deployed as an optimisation layer alongside the existing BMS. The Desigo system continued to control the mechanical plant, while myCoreAI adjusted operating setpoints through 15-minute optimisation cycles based on live building conditions and forecast weather conditions.

Deployment was completed without disruption to building operations and without any requirement to replace existing infrastructure.

Operational carbon reduction

Using UK Government 2025 conversion factors for grid electricity and mains gas, measured savings to date equate to an estimated operational carbon reduction of 29.26 tCO₂e.

At the annualised position reported in the original case study, this equated to approximately 38.4 tCO₂e/year.

Verification note. All reported savings are based on weather-normalised site reporting and verified University of St Andrews meter data, not estimates or simulations.

Leadership commentary

Amanda Cook, BMS Manager

As BMS Manager for a large university estate, I wanted to understand how myCoreAI could be applied within existing building infrastructure. It was implemented at the Scottish Oceans Institute to assess its impact on performance, focusing on strengthening the existing Desigo CC system rather than replacing it.

Since implementation, the results demonstrate measurable improvements in energy use and internal conditions, reinforcing the value of using AI to enhance an existing BMS rather than replace it.

Michal Rosa, BMS Project Engineer

Being a BMS Project Engineer means looking at all stages of planning and deploying new operational technologies. Despite initial pushbacks from the IT Department, the process turned out to be straightforward and scalable.

Allowing AI models to assess our building from a broader perspective and optimise it using external influences such as local weather forecasts, usage patterns and building heat-retention characteristics is an exciting opportunity to drive energy savings and reduce operational carbon emissions.

Next steps

The year-one evidence base can now support rollout to other suitable buildings across the estate, using the same low-disruption deployment approach.

The monthly comparison view provides a clear and accessible way to demonstrate the value delivered by AI optimisation to both operational and financial stakeholders.

PERFORMANCE UPDATE

Subsequent electricity performance continued to strengthen. A later 12-month view (July 2025 - June 2026) showed a 26.3% reduction in HVAC electricity consumption (30.1 MWh), while the September 2025 - June 2026 period showed a 31.5% reduction (29.2 MWh).

EcoAI performance update - outside the original approved April 2025 - April 2026 case-study reporting period.